RACE Regular Analysis Through Continuous Exercise

Continuity : RACE-01

Single Correct Question

1. The function $f(x)=\frac{4-x^2}{4x-x^3}$, is -

(A) discontinuous at only one point in its domain.
(B) discontinuous at two points in its domain.
(C) discontinuous at three points in its domain.
(D) continuous everywhere in its domain.

MCT122

Subjective Type Question

2. A function $y=f(x)$ is defined as $f(x)=\begin{cases} k\sin\frac{(x+3)\pi}{6} & \text{for} \quad x\le 2 \\ \frac{3-\sqrt{11-x}}{x-2} & \text{for} \quad x>2 \end{cases}$. If $f(x)$ is continuous at $x=2$, then find the value of k.

MCT123

3. Discuss the continuity of the function $[\cos x]$ at $x=\frac{\pi}{2}$, where [.] denotes the greatest integer function

MCT124

4. Find whether $f(x)$ is continuous or not at $x=1$, if $f(x)=\begin{cases} \sin\frac{\pi x}{2} & , \quad x<1 \\ [x] & , \quad x\ge 1 \end{cases}$ where [.] denotes the greatest integer function

MCT125

5. Let $f(x)=\begin{cases} x^2 & \text{if} \quad x>2 \\ x+1 & \text{if} \quad x\le 2 \end{cases}$. Discuss continuity of $f(x)$.

MCT126

6. Let $f(x)=\frac{\ell n(1+ax)-\ell n(1-bx)}{x}$. Find the value which should be assigned to f at $x=0$, so that it is continuous at $x=0$.

MCT127

7. If $f(x)=\begin{cases} \frac{x^2}{a} & ; \quad 0\le x<1 \\ -1 & ; \quad 1\le x<\sqrt{2} \\ \frac{2b^2-4b}{x^2} & ; \quad \sqrt{2}\le x<\infty \end{cases}$ then find the value of a & b if $f(x)$ is continuous in $[0,\infty)$

MCT128

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8. Examine the continuity of the function $f(x)=\begin{cases}\dfrac{x^2-1}{x^2-2|x-1|-1}, & x\neq 1\\ \dfrac{1}{2}, & x=1\end{cases}$ at $x=1$.

MCT129

9. Show that the function $f(x)=\begin{cases}2x+3, & -3\le x<-2\\ x+1, & -2\le x<0\\ x+2, & 0\le x\le 1\end{cases}$ is discontinuous at $x=0$ and continuous at other points in the interval $[-3,1]$

MCT130

10. Discuss continuity of the function $f(x)=\dfrac{2x^5-8x^2+11}{x^4+4x^3+8x^2+8x+4}$

MCT131

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ANSWER KEY

1.	D	2.	$k=\frac{1}{3}$	**3.**	discontinuous at $x=\frac{\pi}{2}$	**4.**	continuous at $x=1$
5.	discontinuous at $x=2$			**6.**	$f(0)=a+b$	**7.**	$a=-1$ & $b=1$
8.	Discontinuous			**10.**	continuous everywhere		

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RACE
Regular Analysis Through Continuous Exercise

Continuity : RACE-02

Subjective Type Question

1. If $f(x)$ is continuous in $[a,b]$ then prove that $\exists$ c $\in [a,b]$ such that $f(c)=\frac{2f(a)+3f(b)}{5}$.

MCT132

2. Show that the function $f(x) = x - 1 - \cos\pi x, x \in [1,2]$ takes on the value $1/2$.

MCT133

3. Does the function $f(x) = \frac{x^3}{4} - \sin\pi x + 3$ take on the value $2\frac{1}{3}$ within the interval $[-2,2]$?

MCT134

4. Show that the function $f(x) = (x-a)^2(x-b)^2 + x$ take the value $\frac{a+b}{2}$ for some. value of $x \in [a,b]$.

MCT135

5. Use intermediate value theorem to show that the equation $2x^3 + x^2 - x + 1 = 5$ has a solution in the interval $[1,2]$.

MCT136

6. Let $f(x) = [x]$ & $g(x) = sgn(x)$ (where [.] denotes greatest integer function), then discuss the continuity of $f(x) \pm g(x), f(x).g(x)$ & $\frac{f(x)}{g(x)}$ at $x = 0$.

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MCT137

7. Discuss the continuity of $f(x) = [x] + |x-1|$. (where [.] denotes greatest integer function)

MCT138

8. If $f(x) = [\sin(x-1)] - \{\sin(x-1)\}$ then comment on continuity of $f(x)$ at $x=\frac{\pi}{2}+1$.
(where [.] & {.} denote greatest integer function & fractional part function)

MCT139

9. If $g(x)=\begin{bmatrix} x & \text{if } x\in\mathbb{Q} \\ 0 & \text{if } x\notin\mathbb{Q}\end{bmatrix}$, then find the points where function is continuous.

MCT140

10. If $f(x)=\begin{cases} x^2 & ; \ x\in\mathbb{Q} \\ 1-x^2 & ; \ x\notin\mathbb{Q}\end{cases}$, then find the points where function is continuous.

MCT141

11. Discuss the continuity of the function $f(x)=\begin{cases}\frac{e^{1/(x-1)}-2}{e^{1/(x-1)}+2}, & x\neq 1 \\ 1, & x=1\end{cases}$ at $x = 1$.

MCT056

ANSWER KEY

6. All are discontinuous at $x = 0$.
7. discontinuous at all integral points
8. Discontinuous
9. $x = 0$
10. $x=\pm\frac{1}{\sqrt{2}}$
11. discontinuous

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